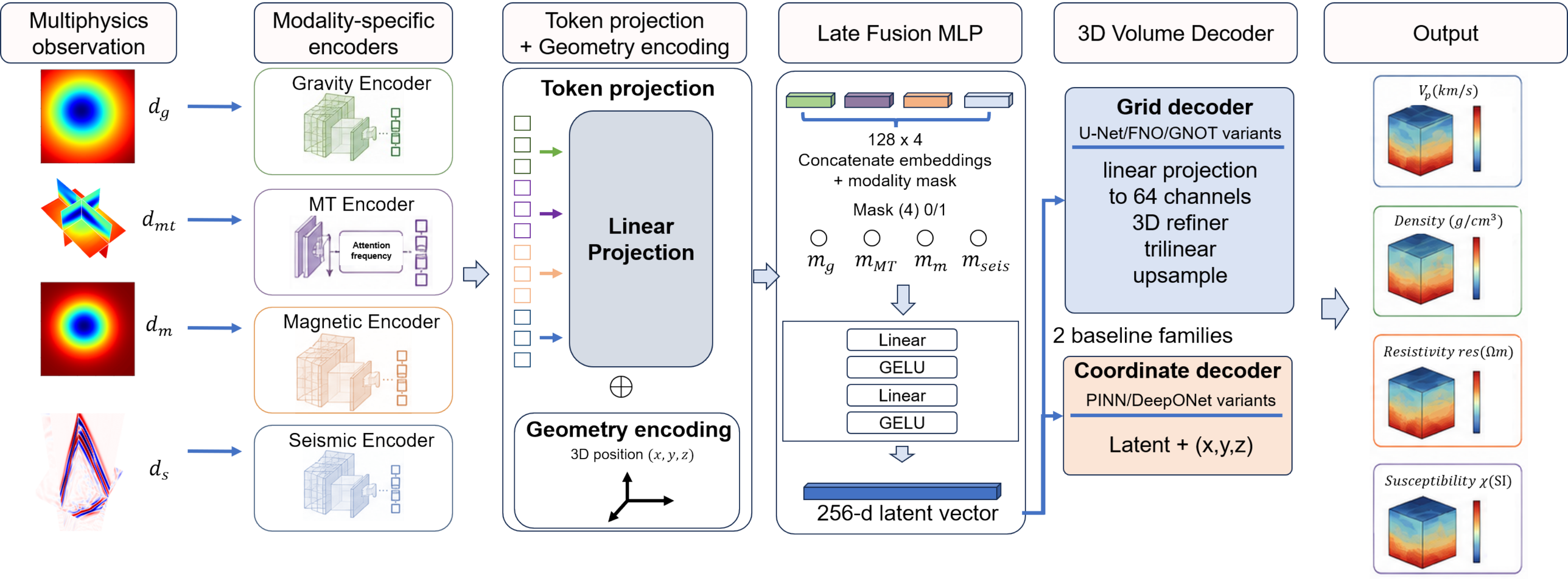

Multiphysics observation
Modality-specific encoders
Token projection + Geometry encoding
Late Fusion MLP
3D Volume Decoder
Output
d_g
d_{mt}
d_m
d_s
Gravity Encoder
MT Encoder
Attention frequency
Magnetic Encoder
Seismic Encoder
Token projection
Linear Projection
Geometry encoding
3D position (x, y, z)
128 x 4
Concatenate embeddings
+ modality mask
Mask (4) 0/1
m_g m_{MT} m_m m_{seis}
Linear
GELU
Linear
GELU
256-d latent vector
Grid decoder
U-Net/FNO/GNOT variants
linear projection
to 64 channels
3D refiner
trilinear
upsample
2 baseline families
Coordinate decoder
PINN/DeepONet variants
Latent + (x,y,z)
$V_p(km/s)$
$Density\ (g/cm^3)$
$Resistivity\ res(\Omega m)$
$Susceptibility\ \chi(\mathrm{SI})$